EDISON OCHIENG

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PROFESSIONAL SUMMARY

Mechanical Engineering graduate from the University of Nairobi, with hands-on CAD experience in SolidWorks, Autodesk Inventor, and Fusion 360, gained during a 12-week industrial attachment at KIRDI. Demonstrated ability to produce 3D models, assemblies, and 2D technical drawings for industrial prototypes. Final-year project involves coupled FE simulation of sheet metal forming processes, reflecting strong manufacturing and design fundamentals. Eager to contribute to product development, prototyping and technical support.

KEY SKILLS

- Manufacturing Processes & DFM Principles
- CAD & Engineering Design – SolidWorks, Inventor, Fusion 360 (3 years hands-on experience)
- Prototype Fabrication & Lab Testing
- Simulation & FEA (SolidWorks Simulation, Abaqus)
- Technical Drawing & Documentation
- Computing & Analytical Tools (Microsoft Office Suite)
- Leadership, Collaboration & Soft Skills

INDUSTRIAL EXPERIENCE

Industrial Attachment – Mechanical Engineering *June – August 2025 (12 weeks)*

Kenya Industrial Research and Development Institute (KIRDI), Nairobi

- Collaborated in a team to design and fabricate Jiko stove components from sheet metal, applying DFM principles and manufacturing tolerances to ensure functional assemblies.
- Operated laser cutting machines to produce sheet-metal components to precise dimensional tolerances, maintaining edge quality.
- Created 3D models, assemblies, and detailed 2D technical drawings using SolidWorks, Autodesk Inventor, and Fusion 360 for industrial prototypes and machinery components.
- Contributed to the design and fabrication of a poultry de-feathering machine, supporting concept development, component iteration, and mechanical assembly.
- Performed machine shop operations including shaft milling and lathe work, developing hands-on competency in material removal processes.

CERTIFICATIONS

- Certified SOLIDWORKS Associate - LinkedIn Learning
- SOLIDWORKS 2024 Essential Training - LinkedIn Learning
- Autodesk Inventor Design Certification - LinkedIn Learning
- Certificate of Participation - Autodesk Designathon 2026
- Certificate of Participation - Autodesk Designathon 2025

ACADEMIC & PERSONAL PROJECTS

Freelance Client Project – EV Battery Cooling Pack

Design and Thermal Simulation of an Electric Vehicle (EV) Battery Cooling Pack

- Designed and simulated a full electric vehicle battery cooling pack for a private client using SolidWorks and Autodesk Inventor, covering 3D modeling, assembly, and thermal performance analysis.
- Iterated cooling pack geometry and layout to keep battery cells within the required operating temperature range.
- Delivered a complete design package, 3D models, assemblies and simulation results and supporting manufacturability and thermal performance requirements.

Final-Year Project *2025 – 2026*

Forming Limit Curve Prediction for AM AlSi10Mg through the GTN–Hill'48 Coupled Simulation of the Nakajima Test

- Applied sheet metal forming simulation (Nakajima test) using coupled damage mechanics, directly relevant to DFM and manufacturability analysis.
- Translates to practical failure prediction for sheet-metal parts before they are manufactured.
- Developed understanding of material behaviour under forming conditions, including failure prediction and process optimization.

EXTRACURRICULAR ACTIVITIES & LEADERSHIP

- Leader, Workshop and Exhibition Planning - Women in Engineering Students Summit 4.0 (WIESS 4.0), University of Nairobi (2025–2026)
- Member, Mechanical Engineering Students Association (MESA), University of Nairobi (2022–2026)
- Member, Engineering Students Association (ESA), University of Nairobi (2022–2026)
- Secretary, Student Council, Aoch Muga Secondary School (2018–2020)

EDUCATION

Bachelor of Science in Mechanical Engineering *2021 – 2026*

University of Nairobi, Kenya

Relevant Coursework: Engineering Design I, II & III, Thermodynamics, Fluid Mechanics, Strength of Materials, Manufacturing Processes, CAD/CAM, Engineering Drawing, Control Systems, Materials Science

Kenya Certificate of Secondary Education (KCSE) *Completed 2020*

Aoch Muga Secondary School, Homa Bay - Mean Grade: A

Kenya Certificate of Primary Education (KCPE) *Completed 2016*

Aoch Muga Primary School, Homa Bay - Scored 347 out of 500

LANGUAGES

English, Kiswahili